

HASEEB AHMAD

AI / ML Engineer

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SUMMARY

Computer Science student and Software Engineer specializing in **AI/ML engineering, agentic systems, and LLM-powered applications**. Experienced in building production-oriented, AI-driven backend systems using Python, FastAPI, LangGraph, and RAG pipelines, with hands-on work across the ML lifecycle from data processing and model integration to deployment. Actively contributing to open-source AI infrastructure projects and expanding practical expertise in MLOps, model deployment pipelines, and scalable AI systems.

EDUCATION

COMSATS University Islamabad – Lahore Campus

2024 – 2028

B.Sc. Computer Science, 4th Semester – CGPA: 3.92 / 4.0

VOLUNTEER EXPERIENCE

Google Developers Groups on Campus (GDGoC)), COMSATS Lahore – Technical Co-Lead

2025 – Present

Technical Lead – developer community, events, and technical initiatives

- Led technical initiatives and coordinated **2+ technical events and workshops**, handling planning, technical execution, and cross-team collaboration.
- Maintained and enhanced the developer community website by implementing new features and improving existing functionality and user experience.
- Designed and implemented an **automatic certificate generation and download system**, reducing manual administrative effort.
- Mentored **50+ students** in Programming Fundamentals and Object-Oriented Programming while promoting hands-on technical learning.

PROJECTS

CodeCrew – Autonomous Multi-Agent Software House

Python, FastAPI, LangGraph, React, PostgreSQL, GitHub API – 2026

Live Demo: <https://codecrew-ashen.vercel.app/>

- Built an autonomous multi-agent software engineering platform that simulates a complete development team for repository analysis, architecture planning, code generation, testing, and technical documentation.
- Designed a stateful **LangGraph** orchestration workflow with specialized Planner, Backend, Frontend, Database, Testing, and Documentation agents collaborating through structured agent state.
- Integrated **GitHub** automation for repository cloning, codebase analysis, feature-branch creation, automated implementation, and generation of verified Pull Requests.
- Implemented fault-tolerant workflow execution using **PostgreSQL/SQLite checkpoints**, JWT/Argon2 authentication, sandboxed repository processing, and automated CI/CD validation using Ruff, Prettier, and TypeScript.

MentorX – AI Academic Intelligence & Admission Guidance Platform

Next.js, FastAPI, LangGraph, Qdrant, PostgreSQL, Gemini, Groq – 2026

Live Demo: <https://mentor-x-beryl.vercel.app/>

- Built an agentic AI academic guidance platform providing data-backed university admission counseling, aggregate calculations, merit analysis, and interdisciplinary degree transition guidance for FSc and O/A-Level students.
- Designed an adaptive **LangGraph RAG pipeline** that retrieves and grades document relevance, conditionally rewrites queries and performs Tavily web searches, then refines context before generating evidence-grounded responses.
- Implemented a hybrid knowledge system using **Qdrant** for semantic retrieval of university prospectuses, policies, and merit documents, with **PostgreSQL and SQLAlchemy** for users, document metadata, and transactional data.
- Developed secure multi-role access with **Google OAuth** and role-based authorization, alongside an admin document ingestion pipeline for PDF extraction, configurable chunking, embedding generation, and vector indexing.

CampuzLift – Intelligent Campus Ride-Sharing Platform

React, Node.js, Express, PostgreSQL, Prisma – 2026

Live Demo: <https://campuz-lift.vercel.app/>

- Built a full-stack campus carpooling platform matching student and faculty drivers with passengers using fixed-route corridor matching.
- Engineered a custom geospatial matching engine to calculate point-to-route distances and verify pickup/drop-off directionality along driver routes.
- Designed a weighted ranking system based on route compatibility, pickup proximity, schedule alignment, and driver ratings.
- Reduced route storage by **70%+** using PostgreSQL caching and Douglas-Peucker geometry simplification while maintaining **10-meter spatial accuracy**.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript

Backend: FastAPI, Node.js, Express.js, Unicorn, RBAC

AI & Agent Systems: LangGraph, LangChain, RAG, Corrective-RAG, LLM Workflows, Agentic AI, Google Gemini, Groq, Embeddings, Feature Engineering

Machine Learning: Scikit-learn, Pandas, NumPy, XGBoost, TensorFlow, Keras, EDA

Databases: PostgreSQL, MongoDB, Qdrant

Frontend: Next.js, React.js, Tailwind CSS

DevOps & Tools: Docker, Git, GitHub Actions, MLOps (currently learning)

Core: OOP, Software Engineering, Technical Documentation

ACHIEVEMENTS & EXTRACURRICULAR

- 1st Place** – 2x Web Hackathon 2026
- 3rd Place** – App Development Competition, SOFTEC '26
- Technical Lead** – Google Developers on Campus (Oct 2025 – Present)
- Mentored 50+ students** in Programming Fundamentals and Object-Oriented Programming (OOP).